

DreamBooth

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Introduction

Dreambooth seeks to solve the problem of being able to generate images of a specific subject without extensive training data of that subject. In the original paper “DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation” by Nataniel Ruiz et al., this problem was solved by fine-tuning existing diffusion models with data of a subject labeled with a unique adjective token, allowing the model to use existing knowledge about how that category of subject, while also retaining unique features of the subject.

Chosen Result

Our goal was to recreate the capability to create novel scenes of a subject using limited training data. This capability was especially intriguing to us because of how efficiently it can accomplish such a challenging task, especially relative to the naive way image generation is often done. How well a subject is recreated is somewhat subjective, so the original paper created metrics to measure subject and prompt fidelity (matching the input images and matching the prompt), which we hoped to be able to compare our results with.

Method	DINO ↑	CLIP-I ↑	CLIP-T ↑
Real Images	0.774	0.885	N/A
DreamBooth (Imagen)	0.696	0.812	0.306
DreamBooth (Stable Diffusion)	0.668	0.803	0.305
Textual Inversion (Stable Diffusion)	0.569	0.780	0.255

Table 1. Subject fidelity (DINO, CLIP-I) and prompt fidelity (CLIP-T, CLIP-T-L) quantitative metric comparison.

Method	Subject Fidelity ↑	Prompt Fidelity ↑
DreamBooth (Stable Diffusion)	68%	81%
Textual Inversion (Stable Diffusion)	22%	12%
Undecided	10%	7%

Table 2. Subject fidelity and prompt fidelity user preference.

Methodology

Our re-implementation is built on Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5), a latent diffusion model composed of four components:

TABLE 1 – STABLE DIFFUSION COMPONENT OVERVIEW

COMPONENT	ROLE	TRAINING STATUS
UNet2DConditionModel	Noise predictor (denoising backbone)	Frozen LoRA adapters trained
AutoencoderKL (VAE)	Encodes/decodes images to/from latent space	Frozen
CLIPTextModel	Encodes text prompts into embeddings	Frozen
DDPMScheduler	Controls the forward noise schedule (α_t , σ_t)	N/A

Only the UNet is updated during training; the VAE and text encoder remain frozen throughout, following the original DreamBooth protocol. If LoRa is enabled for the model, instead of directly changing the UNet parameters, pairs of smaller A and B matrices are updated instead and added to the original weights. We use the official DreamBooth dataset (3–6 images per subject). Prior preservation images (Stream B) are pre-generated via the base SD model and stored before training. All images are resized to 512×512 with center-crop normalization.

Training uses a prior preservation loss from the original paper(Equation 2):

$$L_{DB} = E[||\epsilon_{\theta}(z_t, t, c) - \epsilon||^2] + \lambda \cdot E[||\epsilon_{\theta}(\tilde{z}_t, t, c) - \epsilon||^2]$$

- The first term is subject reconstruction loss (Stream A: instance images with prompt “a [V] dog”).
- The second term is prior preservation loss (Stream B: class images with prompt “dog”). $\lambda=1.0$ by default.

We replicated the three quantitative metrics used in the original paper:

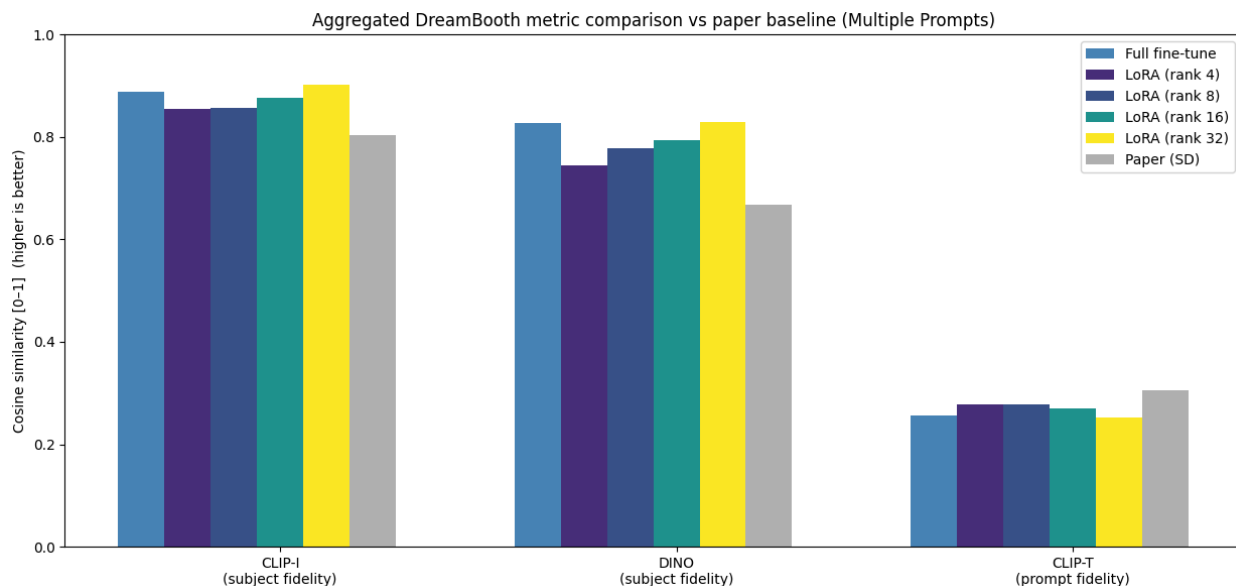
- DINO: Average pairwise cosine similarity between DINO ViT-S/16 CLS token embeddings of the generated images and the real reference images. (Target $\rightarrow$ 0.668)
- CLIP-I: Average pairwise cosine similarity between CLIP ViT-B/32 image embedding of generated and real images. This metric measures subject fidelity at a semantic level. (Target $\rightarrow$ 0.803)
- CLIP-T: Average cosine similarity between CLIP image embeddings (generated) and CLIP text embeddings (evaluation prompt). Measures prompt/context fidelity. (Target $\rightarrow$ 0.305)

Modifications:

- Base Model: We were only able to use the publicly available Stable Diffusion v1.5.
 - The original paper used both Imagen (a proprietary Google model) and Stable Diffusion (likely an older version than we used, but they didn't specify what they used).
- LoRA instead of full fine-tuning: Rather than updating all UNet weights, we inject trainable low-rank adapter matrices A and B into the query, key, value, and output projections of every UNet attention block. The frozen weight W is augmented as $W(x) + BA(x) \cdot (\alpha/r)$, with B initialized to zero. Only A and B are trained ($\sim 0.3M$ vs $\sim 860M$ parameters). We experiment with ranks $r \in \{4, 8, 16, 32\}$.
- We likely used a different unique adjective token to mark our subject. The original paper didn't specify a specific token to use, but simply a process to help identify rare tokens. However, a commonly used token for reimplementations of the paper is “sks” (Hugging Face), which is what we picked, as it has been proven to be a reliable and rare token for this purpose.

Results & Analysis

We tested our model on the same three metrics that the original paper used (DINO, CLIP-I, and CLIP-T). Overall, our results were very satisfactory, usually improving upon the original paper's implementation. For CLIP-I and DINO, our subject fidelity (truthfulness to the input images) results were about 10% better than the original paper, and even the lowest rank of LoRA that we tried improved upon the original paper. On the other hand CLIP-T, our prompt fidelity (truthfulness to the intended text modification of the image) is slightly weaker than that of the original paper. This is expected to some extent, since a closer fidelity to the input subject images must be correlated with a slightly lower fidelity to anything trying to modify that image (the text). Since the improvement in subject fidelity is a lot higher than the decrease in prompt fidelity, we believe this was a worthwhile tradeoff. The Dreambooth authors did not mention which Stable Diffusion model they tested on, but we suspect that some of our improvements may just be an improvement in the base Stable Diffusion model that we fine-tuned on.



As we can see in the graph above, tweaking the rank of LoRA to reduce computation cost seemed, on average, to have a very small effect on fidelity. On the other hand, something not visible in the graph is that lower ranks tended to make the output qualitatively worse in a way that metrics struggled to measure. This difference is not so stark that it's clear from just looking at the Rank 4 images alone, but it's clear when compared to the higher rank images. This is likely just due to the tighter information bottleneck of the Rank 4 LoRA limiting the model's ability to make really subtle modifications to the image, as opposed to a higher rank LoRA or full fine-tuning.

Overall, the ability of our LoRA-tuned version to achieve nearly the same quality performance at a significantly lower training cost seems to validate a larger shift into parameter-efficient fine-tuning algorithms recently in the machine learning space. This will only get more valuable for memory efficiency reasons as models continue to get larger and we discover more useful ways to apply fine-tuning to solve complex problems.

Reflections

We learned that for metrics, LoRA is a good substitute for fine-tuning, as evidenced by the model with the lowest rank of 4 showing only a marginal decrease in subject fidelity metrics compared to fine-tuning, with other LoRA models even doing better. However, we also learned that quantitative metrics don't always correlate with the quality of images produced, as there was a notable difference in quality and subject similarity observed between the lower-ranked LoRa models and the higher-ranked LoRa models and the fine-tuned model. This showed us that it's often not possible to make objective improvements with no downsides. Instead, there are usually tradeoffs and you have to choose what to optimize for when attempting to improve a model, whether it's preserving a subject vs generating more accurate scenes or running efficiently vs with high quality output.

Overall, our reimplementing of the Dreambooth paper was able to achieve competitive results using only publicly available models and parameter-efficient fine-tuning. A key takeaway from this is that proprietary infrastructure is not necessary for effective subject generation, as LoRA and open-source models are sufficient to achieve the desired results.

For future work for this project, it could include exploring how well this method can generalize to learning multiple subjects and more complicated types of visually complex subjects (e.g., humans, scenes with fine-grained detail) to increase its utility. Additionally, learning how LoRA rank scales with subject complexity can help us understand the concrete limitations of LoRA for subject generation. Also, adjusting the learning rates will also be valuable to know what the optimal one is for both the fine-tune and LoRa variations of our model.

References

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